

1. INTRODUCTION

- Computer-Aided Design (CAD) model generation typically requires domain expertise, specialized software, and substantial manual effort.
- Multimodal large language models (LLMs) can process diverse input modalities and have the potential to assist with repetitive CAD tasks while lowering the barrier to entry for non-expert users.
- Our previous 2D CAD generation study showed a performance ordering Text < Image < Vector across input modalities.
- This study investigates whether the same ordering is preserved in 3D CAD generation and examines effective input representations for LLM-based 3D CAD generation.

2. METHODS

2.1. Overall Framework

- Inputs from each modality are converted into executable CAD code, and the resulting 3D model is evaluated against the ground-truth geometry using pose-aligned Voxel Intersection over Union (Voxel IoU).
- The LLM generates replicad JavaScript code, with geometric operations performed through the replicad API, which dispatches to the Open CASCADE Technology (OCCT) kernel.

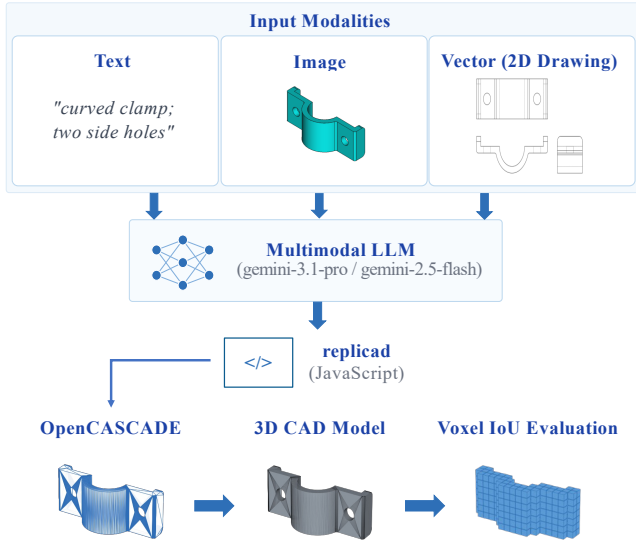


Fig 1. 3D CAD Generation and Evaluation Pipeline

2.2. Input Modality Definitions

Table 1. Input Configuration by Modality

Input	Input Information	Key Characteristics
Text	Natural-language part description	Part type and major geometric features
Image	Single isometric rendering	Overall shape and relative feature positions
Vector	Orthographic drawing in Scalable Vector Graphics (SVG) format	Coordinate-based geometric information from front, top, and side views

2.3 Evaluation Dataset

- The original Standard for the Exchange of Product Model Data (STEP) models of 20 mechanical parts selected from a public dataset were used as the ground-truth geometry.
- For all three input conditions (Text, Image, and Vector), the input representations provided by the dataset were used.

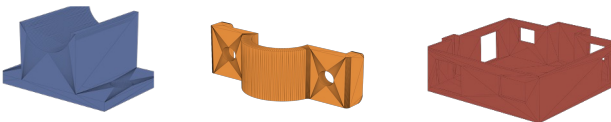


Fig 2. Representative CAD Geometries Used in the Evaluation

3. RESULTS & DISCUSSION

3.1. 3D CAD Generation Performance by Input Modality

- Performance was evaluated on 20 mechanical parts per input modality.
- For gemini-3.1-Pro, mean Voxel IoU was 0.280 for Text, 0.380 for Image, and 0.673 for Vector.
- The ordering of Text < Image < Vector was consistent across both models.
- Vector inputs with explicit coordinate-based geometric information achieved the highest geometric similarity.

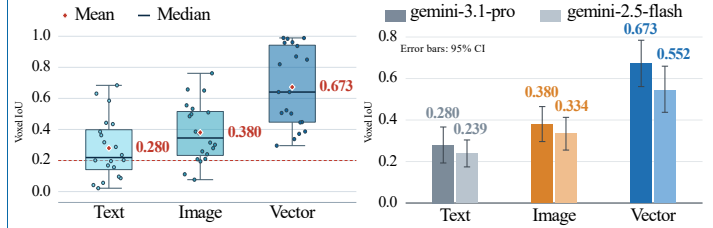


Fig 3. Distribution of Voxel IoU across Input Modalities

Fig 4. Comparison of Input Modality Performance across LLMs

3.2. Comparison of 2D and 3D CAD Generation

Table 2. Comparison of 2D and 3D CAD Generation Performance

Experiment	Text	Image	Vector
2D CAD	0.686	0.728	0.835
3D CAD	0.280	0.380	0.673

- Although lower numerical IoU values were observed in 3D, the performance ordering of Text < Image < Vector was preserved.

3.3. Effects of Geometric Information Content on 3D CAD Generation

- Vector input includes orthographic views and coordinate information from SVG, allowing their effects to be analyzed separately using gemini-2.5-flash.
- To investigate the source of its performance gain, additional tests examined the effects of coordinate removal and the number of image views.
- Removing coordinates reduced mean IoU from 0.55 to 0.41, while increasing the number of image views from 1 to 3 did not improve performance (0.33 → 0.28).
- These results suggest that, under the tested conditions, explicit geometric information contributes more to 3D CAD generation performance than simply increasing the number of views.

Table 3. Experimental Conditions for Geometric Information Analysis

Input	Mean IoU	Information Provided
Vector (SVG)	0.55	Orthographic views + coordinates
Orthographic Images	0.41	Orthographic views
Image (1 View)	0.33	Single-view visual information
Image (3 Views)	0.28	Multi-view visual information

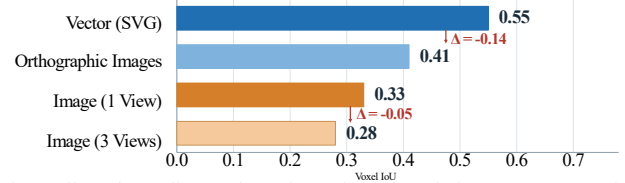


Fig 5. Effect of Coordinate Information and Number of Views on Mean Voxel IoU

4. CONCLUSIONS

Conclusions

- In 3D CAD generation, mean Voxel IoU followed the trend Text < Image < Vector.
- Removing coordinate information from the vector input reduced performance, whereas increasing the number of image views did not improve performance.
- This indicates that explicit geometric information had a greater impact on performance than the number of image views.
- In this study, structured geometric information was more effective for CAD generation than natural-language or image-only inputs.

Future Work

- Future work will investigate a stepwise pipeline that converts unstructured images into orthographic views and derives structured vector representations